Employee Management System
Data Engineering Architecture

Tools & Skills Used

SQL
- Create Tables
- Insert data
- Query data
- Manage data

Linux
- Navigate files
- Basic commands
- Manage files

AWS Services
- S3 (Storage)
- Glue Crawler
- Glue Data Catalog
- Glue Visual ETL
- IAM (Permissions)
- Parquet (Output format)

1. Employee Data (SQL Database)

Create and manage employee data using SQL

employees
employee_id (PK)
name
department_id (FK)
salary
city

departments
department_id (PK)
department_name
manager

IAM Role

AWS Glue service Role Student Project
- Gives Glue Permission to read from S3
- Access Data Catalog
- Write output to S3
- Required for Crawler and ETL job

2. Source Data - Amazon S3 (Raw)

Export the employee data from SQL to a CSV file and store it in an S3 bucket (raw data).
Example Path: s3://my-bucket/raw/employees.csv

3. AWS Glue Crawler

Scan the CSV file in S3, detect the schema (columns and data types) and create/update the meta data in the Glue Data Catalog.

4. AWS Glue Data Catalog

Creates a database (employees_db) and a table (employees) with the schema information and S3 location

5. AWS Glue Visual ETL (Job)

Read data from the Data Catalog (employees table), perform transformations (eg, clean, filter, modify) and write the processed data to S3 in Parquet format

6. Target Data - Amazon S3 (Curated)

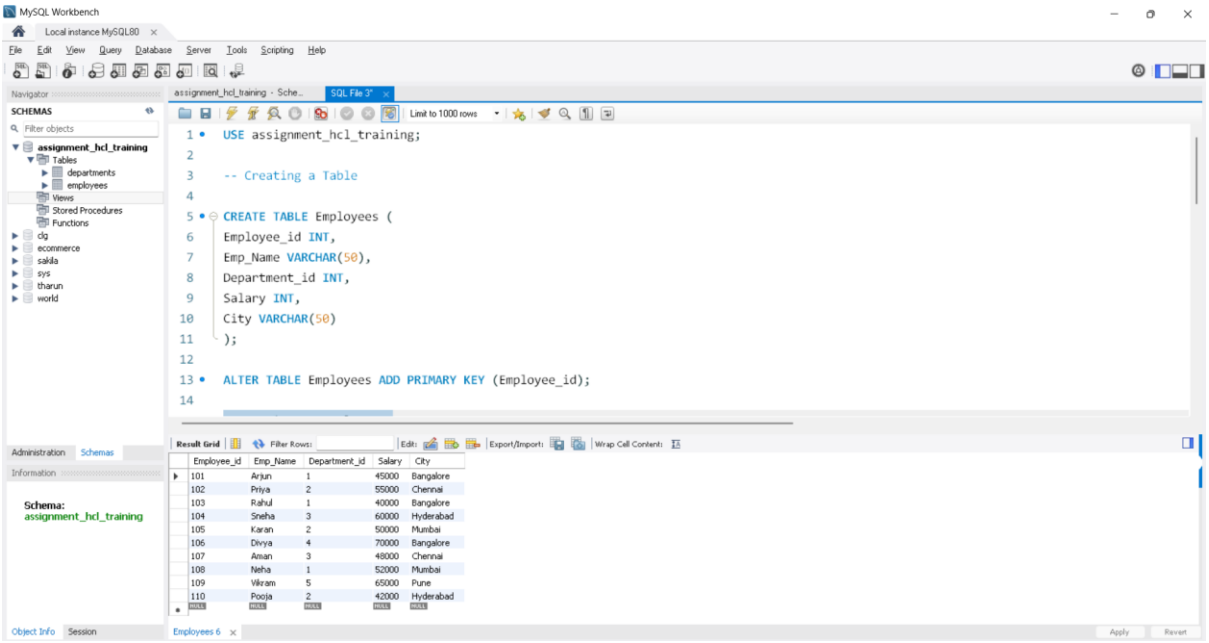
Store the final, transformed employee data in the curated/ folder in Parquet format.

Example path:

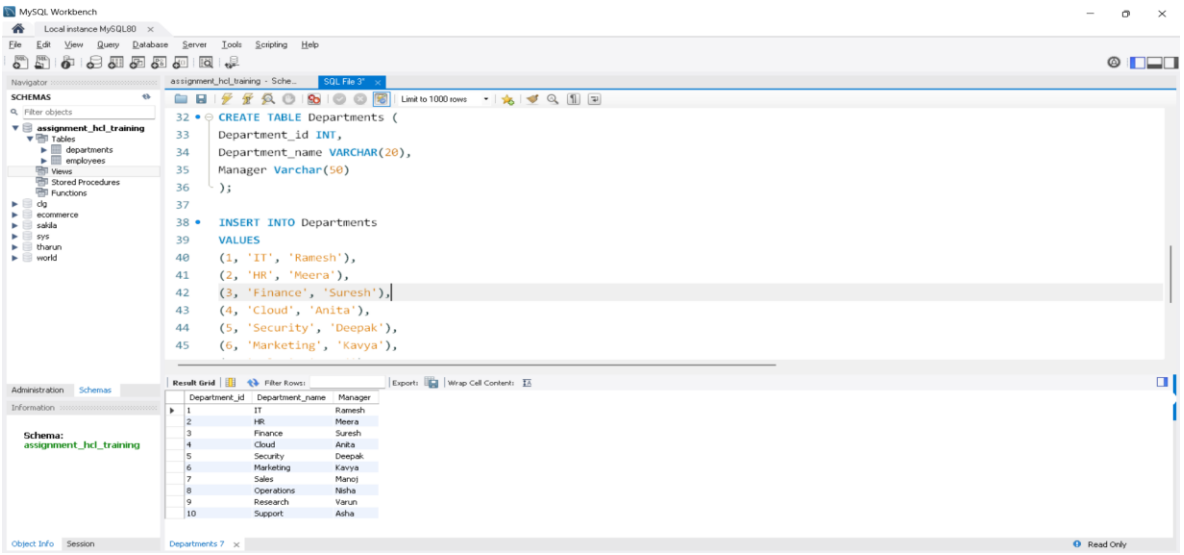
s3://my-bucket/curated/

SQL PRACTICE

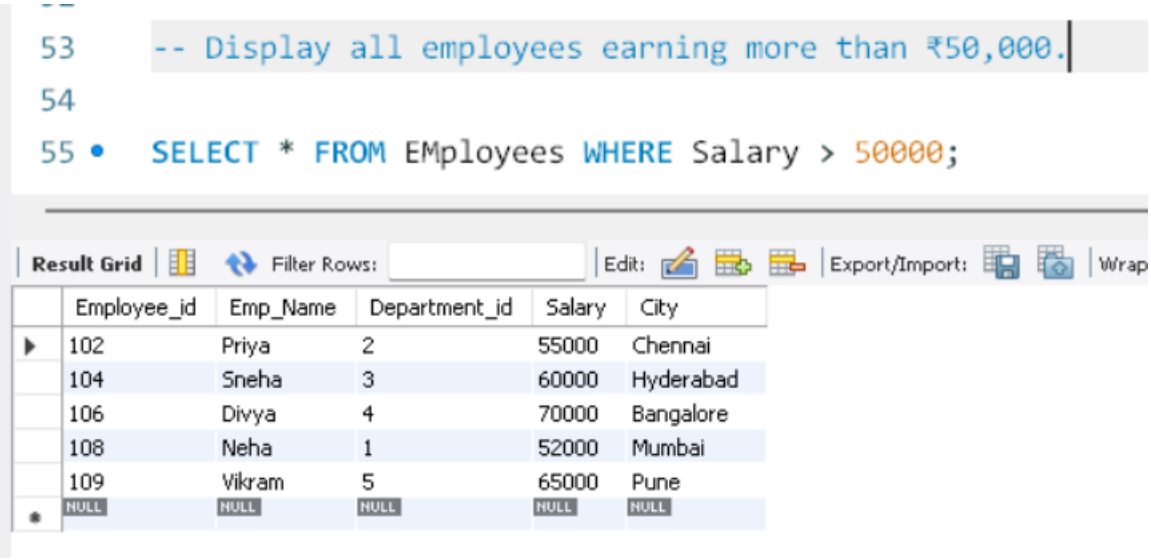
· Created Table Employees



· Created Table Departments



1. Display all employees earning more than ₹50,000.



2. Find employees working in Bangalore.

```
57    -- Find employees working in Bangalore.
58
59 •  SELECT * FROM Employees WHERE City = 'Bangalore';
60
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Employee_id	Emp_Name	Department_id	Salary	City
101	Arjun	1	45000	Bangalore
103	Rahul	1	40000	Bangalore
106	Divya	4	70000	Bangalore
NULL	NULL	NULL	NULL	NULL

3. Display employee names alongside their department names using an INNER JOIN

```
61    -- Display employee names alongside their department names using an INNER JOIN
62
63 •  SELECT E.Emp_name,D.Department_name FROM Employees E INNER JOIN Departments D ON E.Department_id = D.Department_id;
64
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Emp_name	Department_name
Neha	IT
Rahul	IT
Arjun	IT
Pooja	HR
Karan	HR

4. Calculate the average salary in each department.

```
67 •  SELECT  AVG(E.Salary) AS Avg_Salary,D.Department_name
68    FROM Employees E JOIN Departments D
69    ON E.Department_id = D.Department_id
70    GROUP BY D.Department_name;
71
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Avg_Salary	Department_name
45666.6667	IT
49000.0000	HR
54000.0000	Finance
70000.0000	Cloud
65000.0000	Security

5. Find the employee receiving the highest salary.

```
72    -- Find the employee receiving the highest salary.
73
74 •  SELECT Emp_Name,Salary FROM Employees ORDER BY Salary DESC LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Emp_Name	Salary
Divya	70000

6. Count the employees in each department.

```
78 •  SELECT COUNT(E.Employee_id) AS Total_emp,D.Department_name FROM
79    Employees E LEFT JOIN Departments D
80    ON E.Department_id = D.Department_id
81    GROUP BY D.Department_name;
82
83
84
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_emp	Department_name
3	IT
3	HR
2	Finance
1	Cloud
1	Security

7. Display departments that have no employees using a LEFT JOIN.

```
85 •  SELECT D.Department_name FROM
86    Departments D LEFT JOIN Employees E
87    ON E.Department_id = D.Department_id
88    WHERE E.Department_id IS NULL;
89
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_name
Marketing
Sales
Operations
Research
Support

8. Find employees earning more than the average salary.

```
90    -- Find employees earning more than the average salary.
91
92 •  SELECT Emp_name,Salary FROM Employees
93    WHERE
94    Salary > (SELECT AVG(Salary) FROM Employees);
95
96
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Emp_name	Salary
Priya	55000
Sneha	60000
Divya	70000
Vikram	65000

9. Display the top three highest-paid employees.

```
96    -- Display the top three highest-paid employees.
97
98 •  SELECT Emp_name,Salary FROM Employees
99    ORDER BY Salary DESC LIMIT 3;
100
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

Emp_name	Salary
Divya	70000
Vikram	65000
Sneha	60000

10. Find the department with the highest average salary

```
103 •  SELECT D.Department_name,AVG(E.Salary) AS Highest_Avg_salary
104    FROM Departments D JOIN Employees E
105    ON E.Department_id = D.Department_id
106    GROUP BY D.Department_name
107    ORDER BY Highest_Avg_Salary DESC
108    LIMIT 1;
109
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Department_name	Highest_Avg_salary
Cloud	70000.0000

11.Find Employee Name Starting from Letter A

```sql
-- Find Employee Name Starting from Letter A

SELECT Emp_name FROM Employees
HAVING Emp_name LIKE  'A%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Emp_name
Arjun
Aman

12. Display department names and the number of employees in departments that have at least 2 employees.

```sql
-- Display department names and the number of employees in departments that have at least 2 employees.

SELECT D.Department_name,Count(E.Employee_id)
AS No_Of_Emp
FROM Departments D LEFT JOIN Employees E
ON E.Department_id = D.Department_id
GROUP BY D.Department_name
HAVING COUNT(E.Employee_id) >=2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_name	No_Of_Emp
IT	3
HR	3
Finance	2

13. Find departments whose total salary expenditure exceeds ₹100,000.

```sql
-- Find departments whose total salary expenditure exceeds ₹100,000.

SELECT D.Department_name,SUM(E.Salary)
AS Total_salary
FROM Departments D JOIN Employees E
ON D.Department_id = E.Department_id
GROUP  BY D.Department_id,D.Department_name
HAVING SUM(E.Salary) > 100000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_name	Total_salary
IT	137000
HR	147000
Finance	108000

14. Find employees who earn more than the average salary of their own department.

```sql
--  Find employees who earn more than the average salary of their own department.

SELECT Emp_name, Salary, Department_id
FROM Employees E
WHERE Salary > (
    SELECT AVG(Salary)
    FROM Employees
    WHERE Department_id = E.Department_id
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Emp_name	Salary	Department_id
Priya	55000	2
Sneha	60000	3
Karan	50000	2
Neha	52000	1

15. Find cities with at least 2 employees. Display the city name, employee count, and highest salary, sorted by employee

count in descending order.

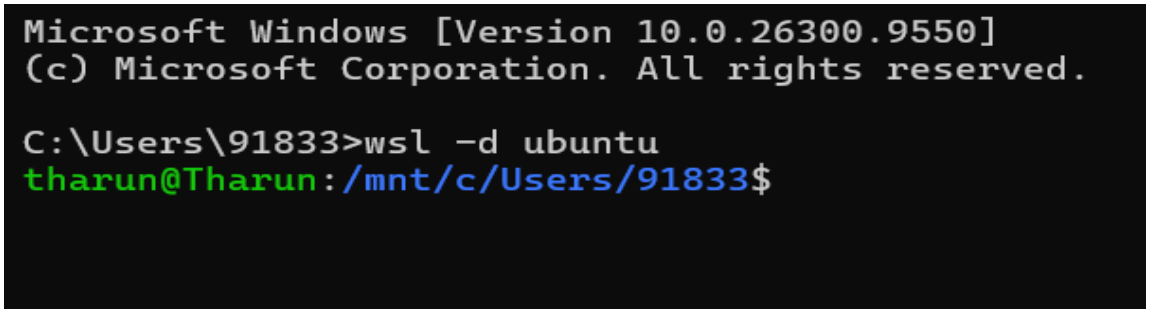
```
143  /*Find cities with at least 2 employees. Display the city name,
144  employee count, and highest salary, sorted by employee count in descending order.
145  */
146
147  SELECT City,
148         COUNT(*) AS No_Of_Emp,
149         MAX(Salary) AS Highest_Salary
150  FROM Employees
151  GROUP BY City
152  HAVING COUNT(*) >= 2
153  ORDER BY No_Of_Emp DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	No_Of_Emp	Highest_Salary
Bangalore	3	70000
Chennai	2	55000
Hyderabad	2	60000
Mumbai	2	52000

LINUX PRACTICE

· Opening Ubuntu



1. **pwd** - Display the current Directory



2.**whoami -** Display the Current username



3.**date -** Display the current date and time



4.**uname -a –** Display System Information

```
tharun@Tharun:/mnt/c/Users/91833$ uname -a
Linux Tharun 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec  1 20:46:23 UTC 2025 x86_64 GNU/Linux
```

5.**hostname** – Display the Computer's Name

```
tharun@Tharun:/mnt/c/Users/91833$ hostname
Tharun
```

6.**ls** – List Files and Directories

```
tharun@Tharun:/mnt/c/Users/91833$ ls
'3D Objects'
AppData
'Application Data'
Contacts
Cookies
Documents
Downloads
Favorites
Links
'Local Settings'
Music
'My Documents'
NTUSER.DAT
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.0.regtrans-ms
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.0.regtrans-ms.cnpf
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.1.regtrans-ms
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.1.regtrans-ms.cnpf
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.2.regtrans-ms
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.2.regtrans-ms.cnpf
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.blf
NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.blf.cnpf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf.cnpf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms.cnpf
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms
NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms.cnpf
NetHood
OneDrive
Pictures
PrintHood
PycharmProjects
Recent
'Saved Games'
Searches
SendTo
'Start Menu'
Templates
Tracing
```

```
NetHood
OneDrive
Pictures
PrintHood
PycharmProjects
Recent
'Saved Games'
Searches
SendTo
'Start Menu'
Templates
Tracing
Untitled-1.c
Untitled.ipynb
Untitled1.ipynb
Untitled2.ipynb
Untitled3.ipynb
Untitled4.ipynb
Videos
ex-list
logs
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini
todo-app
tharun@Tharun:/mnt/c/Users/91833$
```

7. ls -l - Display the files with detailed Information

```
tharun@Tharun:/mnt/c/Users/91833$ ls -l
total 26072
drwxrwxrwx 1 tharun tharun     4096 Jun 24  2021  '3D Objects'
drwxrwxrwx 1 tharun tharun     4096 Oct  5 11:40  AppData
lrwxrwxrwx 1 tharun tharun       34 Oct  4 21:17 'Application Data' -> /mnt/c/Users/91833/AppData/Roaming
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Contacts
lrwxrwxrwx 1 tharun tharun       62 Oct  4 21:17  Cookies -> /mnt/c/Users/91833/AppData/Local/Microsoft/Windows/INetCookies
drwxrwxrwx 1 tharun tharun     4096 Oct  8 05:56  Documents
drwxrwxrwx 1 tharun tharun     4096 Oct 11 08:49  Downloads
drwxrwxrwx 1 tharun tharun     4096 Oct  5 11:40  Favorites
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Links
lrwxrwxrwx 1 tharun tharun       32 Oct  4 21:17 'Local Settings' -> /mnt/c/Users/91833/AppData/Local
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Music
lrwxrwxrwx 1 tharun tharun       28 Oct  4 21:17 'My Documents' -> /mnt/c/Users/91833/Documents
-rwxrwxrwx 1 tharun tharun 15728640 Oct  6 08:39  NTUSER.DAT
-rwxrwxrwx 1 tharun tharun  1118208 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.0.regtrans-ms
-rwxrwxrwx 1 tharun tharun    12288 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.0.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun  1118208 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.1.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.1.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun  1118208 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.2.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.2.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun    73728 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.blf
-rwxrwxrwx 1 tharun tharun    45056 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.blf.cnpf
-rwxrwxrwx 1 tharun tharun    73728 Oct  4 21:17  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf
-rwxrwxrwx 1 tharun tharun     4096 Oct  6 21:27  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf.cnpf
-rwxrwxrwx 1 tharun tharun   561152 Oct  4 21:17  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  6 21:27  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun   561152 Oct  4 21:17  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  6 21:27  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms.cnpf
lrwxrwxrwx 1 tharun tharun       70 Oct  4 21:17  NetHood -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Network Shortcuts
drwxrwxrwx 1 tharun tharun     4096 Oct 10 13:07  OneDrive
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Pictures
lrwxrwxrwx 1 tharun tharun       70 Oct  4 21:17  PrintHood -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Printer Shortcuts
drwxrwxrwx 1 tharun tharun     4096 Jul  5  2022  PycharmProjects
lrwxrwxrwx 1 tharun tharun       59 Oct  4 21:17  Recent -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Recent
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  'Saved Games'
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Searches
lrwxrwxrwx 1 tharun tharun       59 Oct  4 21:17  SendTo -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/SendTo
lrwxrwxrwx 1 tharun tharun       63 Oct  4 21:17 'Start Menu' -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Start Menu
lrwxrwxrwx 1 tharun tharun       62 Oct  4 21:17  Templates -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Templates
```

8.**ls -a** -Display the Hidden Files

```
tharun@Tharun:/mnt/c/Users/91833$ ls -l
total 26072
drwxrwxrwx 1 tharun tharun     4096 Jun 24  2021 '3D Objects'
drwxrwxrwx 1 tharun tharun     4096 Oct  5 11:40  AppData
lrwxrwxrwx 1 tharun tharun       34 Oct  4 21:17 'Application Data' -> /mnt/c/Users/91833/AppData/Roaming
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Contacts
lrwxrwxrwx 1 tharun tharun       62 Oct  4 21:17  Cookies -> /mnt/c/Users/91833/AppData/Local/Microsoft/Windows/INetCookies
drwxrwxrwx 1 tharun tharun     4096 Oct  8 05:56  Documents
drwxrwxrwx 1 tharun tharun     4096 Oct 11 08:49  Downloads
drwxrwxrwx 1 tharun tharun     4096 Oct  5 11:40  Favorites
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Links
lrwxrwxrwx 1 tharun tharun       32 Oct  4 21:17 'Local Settings' -> /mnt/c/Users/91833/AppData/Local
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Music
lrwxrwxrwx 1 tharun tharun       28 Oct  4 21:17 'My Documents' -> /mnt/c/Users/91833/Documents
-rwxrwxrwx 1 tharun tharun 15728640 Oct  6 08:39  NTUSER.DAT
-rwxrwxrwx 1 tharun tharun  1118208 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.0.regtrans-ms
-rwxrwxrwx 1 tharun tharun    12288 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.0.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun  1118208 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.1.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.1.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun  1118208 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.2.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.2.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun    73728 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.blf
-rwxrwxrwx 1 tharun tharun    45056 Oct  7 17:56  NTUSER.DAT{2ad838bb-efea-11ee-a54d-000d3a94eaa1}.TxR.blf.cnpf
-rwxrwxrwx 1 tharun tharun    73728 Oct  4 21:17  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf
-rwxrwxrwx 1 tharun tharun     4096 Oct  6 21:27  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf.cnpf
-rwxrwxrwx 1 tharun tharun   561152 Oct  4 21:17  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  6 21:27  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms.cnpf
-rwxrwxrwx 1 tharun tharun   561152 Oct  4 21:17  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms
-rwxrwxrwx 1 tharun tharun     4096 Oct  6 21:27  NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms.cnpf
lrwxrwxrwx 1 tharun tharun       70 Oct  4 21:17  NetHood -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Network Shortcuts
drwxrwxrwx 1 tharun tharun     4096 Oct 10 13:07  OneDrive
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Pictures
lrwxrwxrwx 1 tharun tharun       70 Oct  4 21:17  PrintHood -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Printer Shortcuts
drwxrwxrwx 1 tharun tharun     4096 Jul  5  2022  PycharmProjects
lrwxrwxrwx 1 tharun tharun       59 Oct  4 21:17  Recent -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Recent
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20 'Saved Games'
drwxrwxrwx 1 tharun tharun     4096 Oct  4 22:20  Searches
lrwxrwxrwx 1 tharun tharun       59 Oct  4 21:17  SendTo -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/SendTo
lrwxrwxrwx 1 tharun tharun       63 Oct  4 21:17 'Start Menu' -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Start Menu
lrwxrwxrwx 1 tharun tharun       62 Oct  4 21:17  Templates -> /mnt/c/Users/91833/AppData/Roaming/Microsoft/Windows/Templates
drwxrwxrwx 1 tharun tharun     4096 Jul  3  2021  Tracing
```

9.**clear** – Clear the Terminal Screen

```
tharun@Tharun:/mnt/c/Users/91833$ clear
```

10.**df-h** – Display disk space usage

```
tharun@Tharun:/mnt/c/Users/91833$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            1.5G     0  1.5G   0% /usr/lib/modules/6.6.114.1-microsoft-standard-
none            1.5G  4.0K  1.5G   1% /mnt/wsl
drivers         238G  144G   94G  61% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.4G  955G   1% /
none            1.5G   36K  1.5G   1% /mnt/wslg
none            1.5G     0  1.5G   0% /usr/lib/wsl/lib
rootfs          1.5G  2.8M  1.4G   1% /init
none            1.5G  492K  1.5G   1% /run
none            1.5G     0  1.5G   0% /run/lock
none            1.5G     0  1.5G   0% /run/shm
none            1.5G  100K  1.5G   1% /mnt/wslg/versions.txt
none            1.5G  100K  1.5G   1% /mnt/wslg/doc
C:\             238G  144G   94G  61% /mnt/c
D:\             932G  277G  656G  30% /mnt/d
tmpfs           1.5G     0  1.5G   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           288M   12K  288M   1% /run/user/1000
tharun@Tharun:/mnt/c/Users/91833$
```

11. **Mkdir -p** – Create a Directory

```
tharun@Tharun:/mnt/c/Users/91833$ mkdir -p LinuxPractice
tharun@Tharun:/mnt/c/Users/91833$
```

12.**cd**- Change Directory

```
tharun@Tharun:/mnt/c/Users/91833$ cd LinuxPractice
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$
```

13.**touch** – Create a Empty File

```
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$ touch notes.txt
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$
```

14.**echo** – Write text into file

```
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$ echo "Linux Commands Practice" > notes.txt
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$
```

15.**cat** – Display the file contents

```
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$ cat notes.txt
Linux Commands Practice
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$
```

16. **Cp –** copy the file

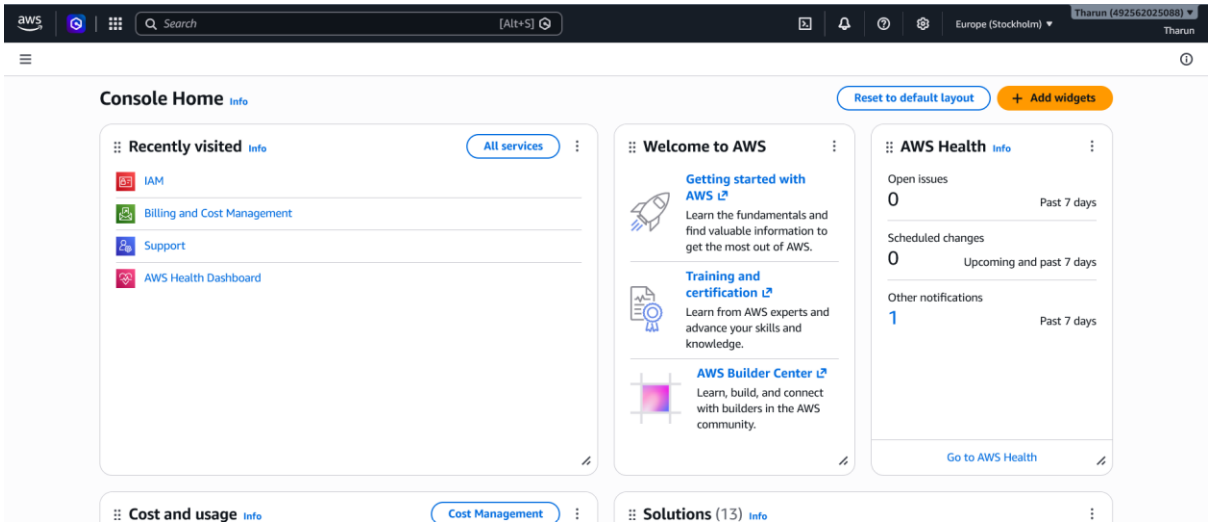
```
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$ cp notes.txt backup.txt
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$
```

17.**mv –** Rename the copied file

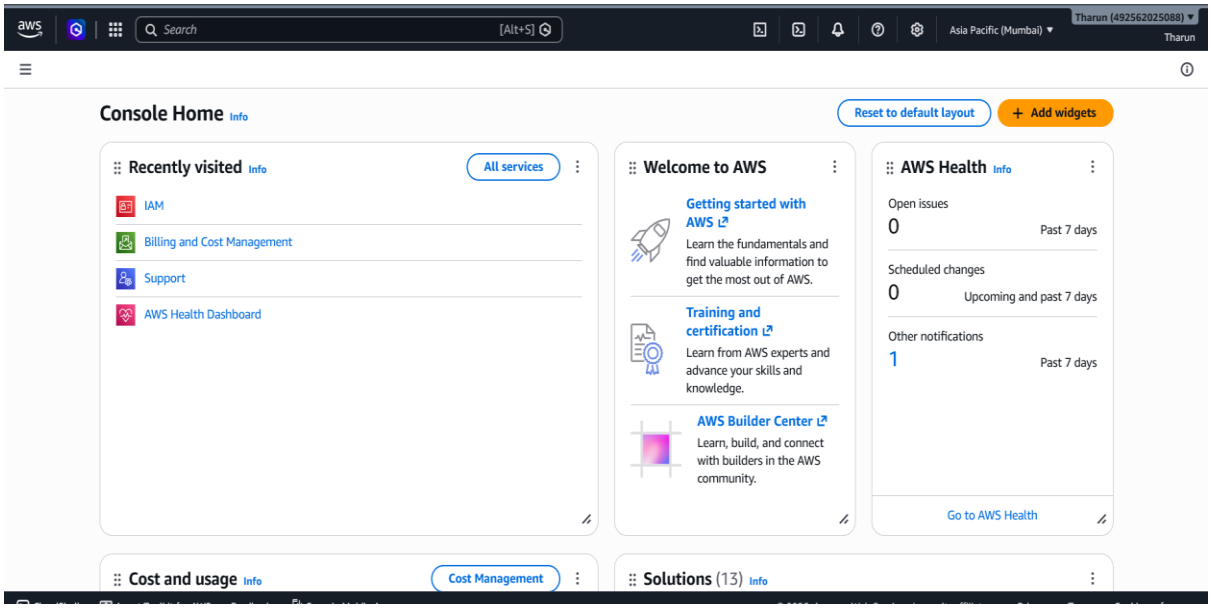
```
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$ mv backup.txt copied_notes.txt
tharun@Tharun:/mnt/c/Users/91833/LinuxPractice$
```

AWS ACCOUNT / REGION

·AWS Console

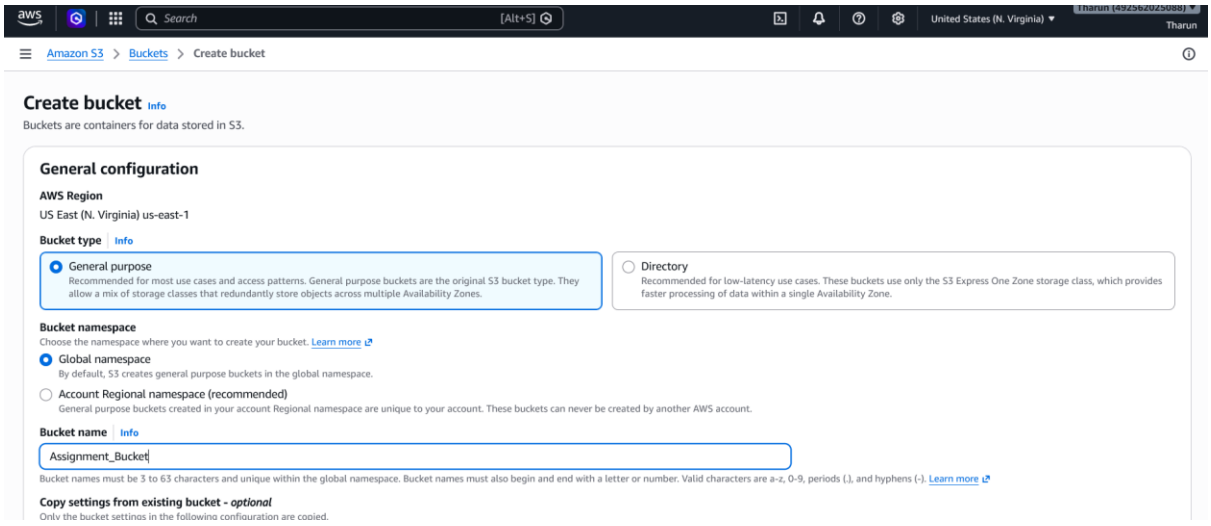


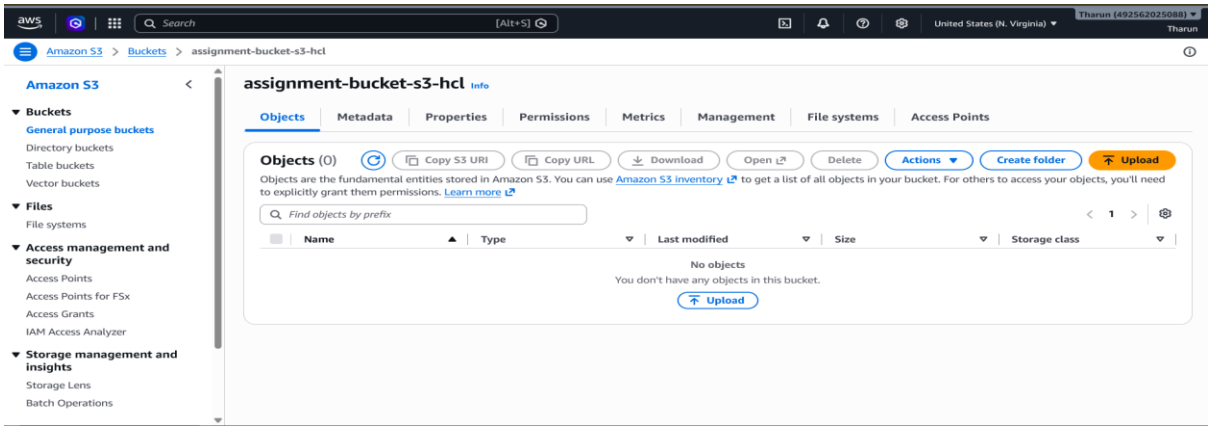
2. Selected AWS Region



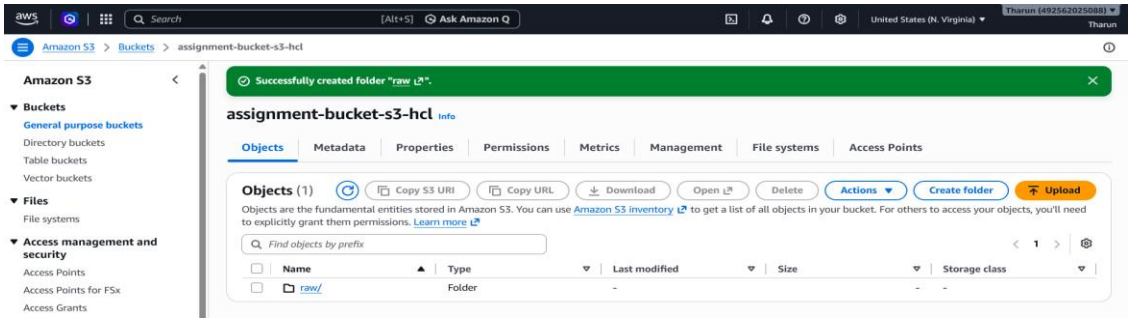
AMAZON S3

1. S3 bucket created

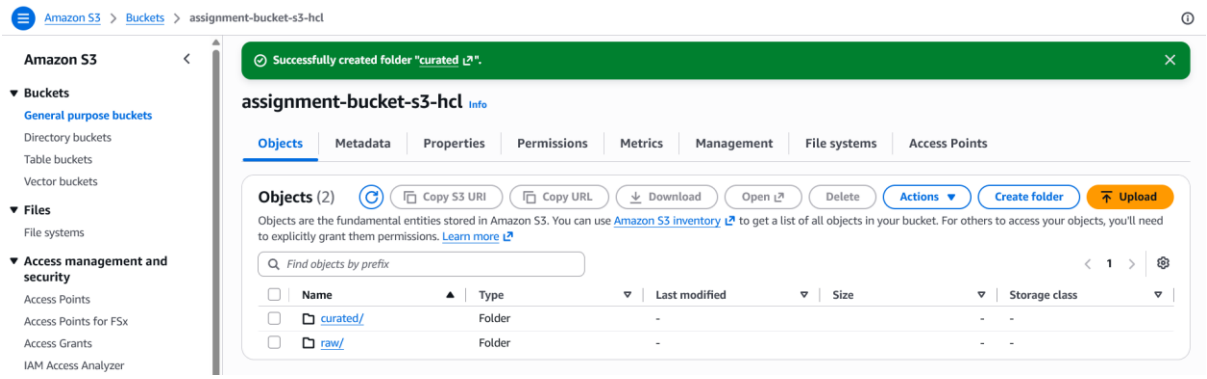




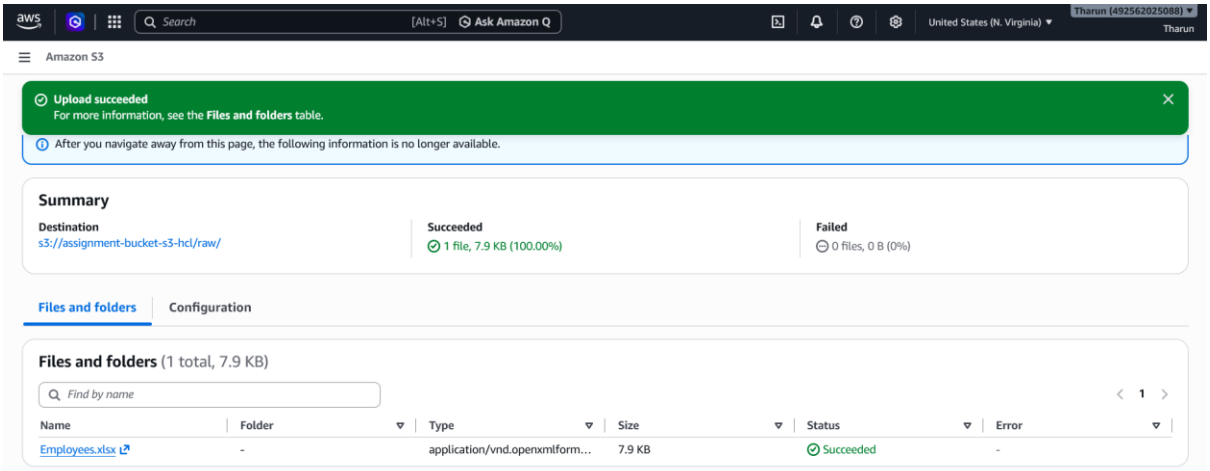
1. Created raw Folder

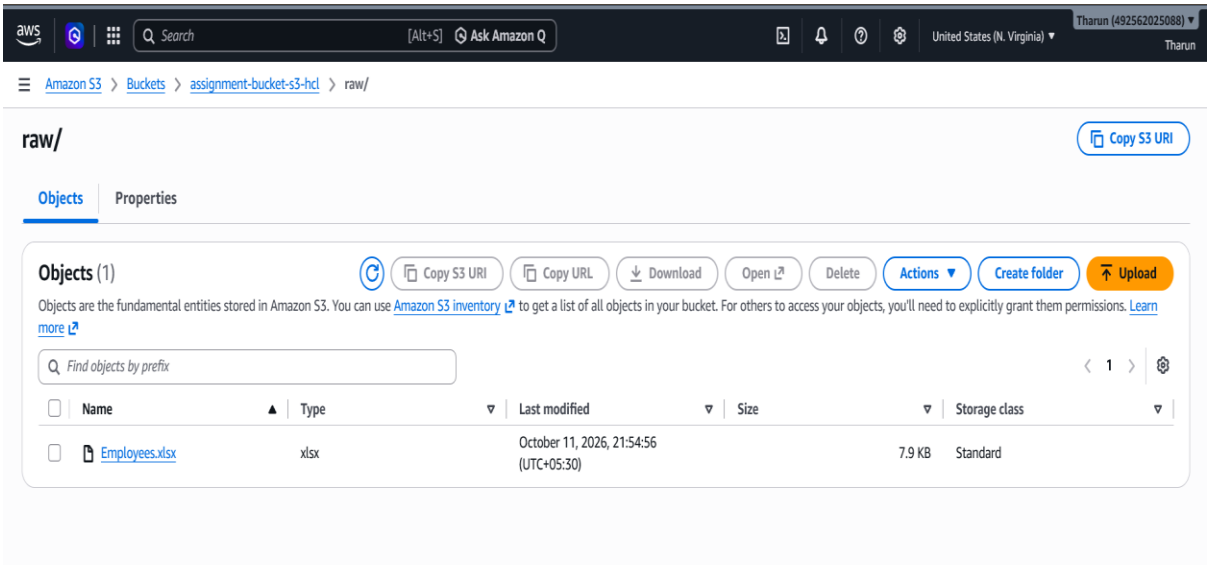


2.Created Curated Folder



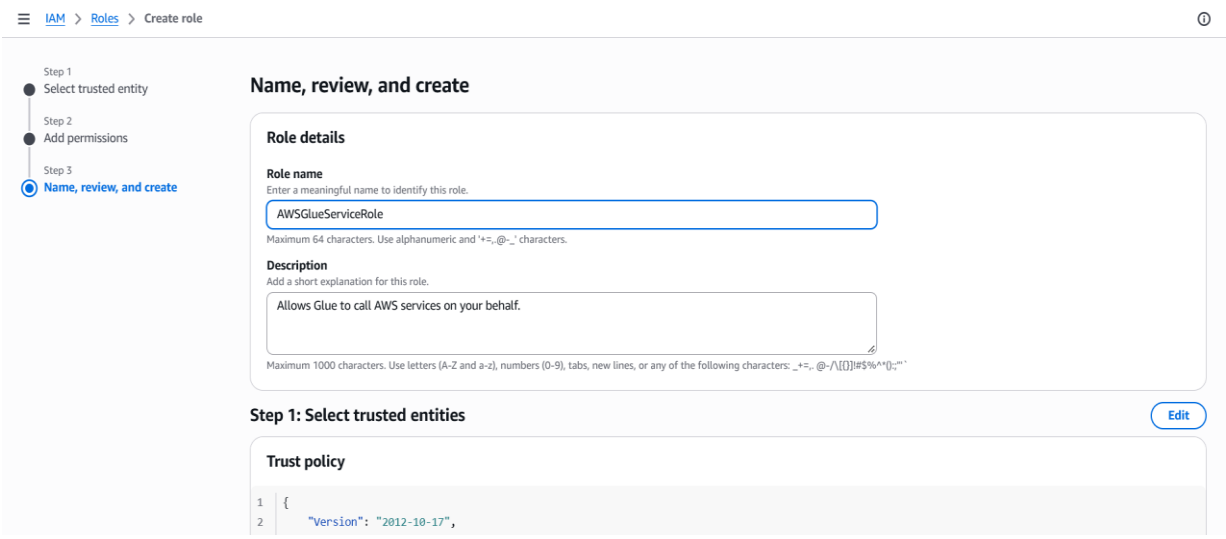
3. employees.csv uploaded inside raw/

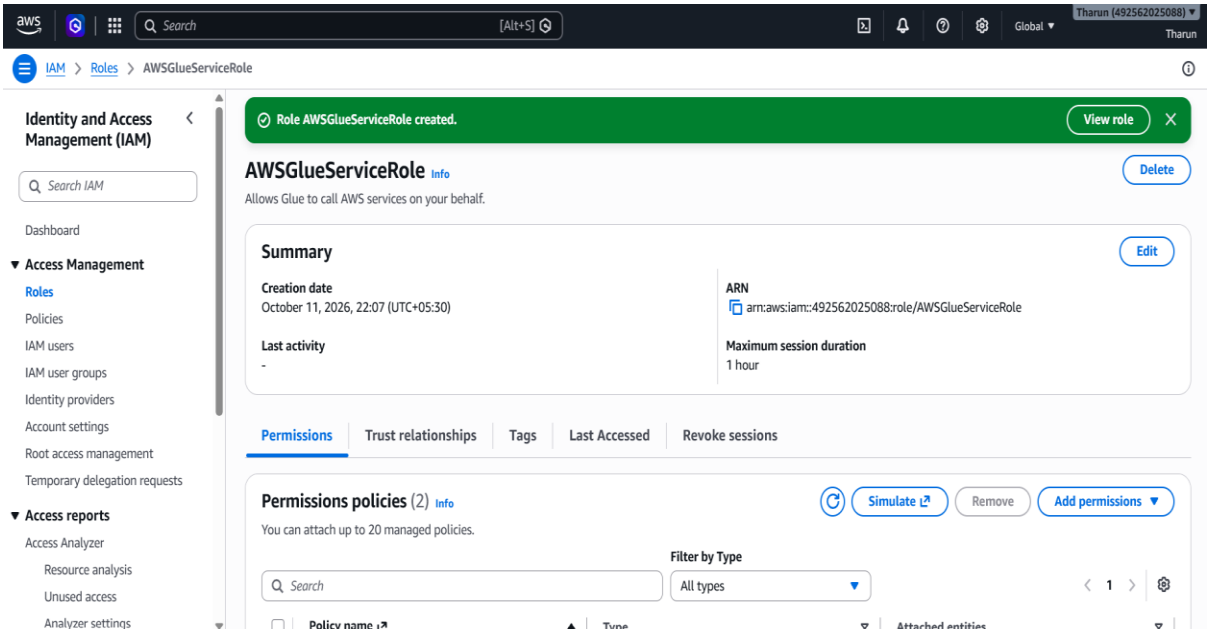




IAM

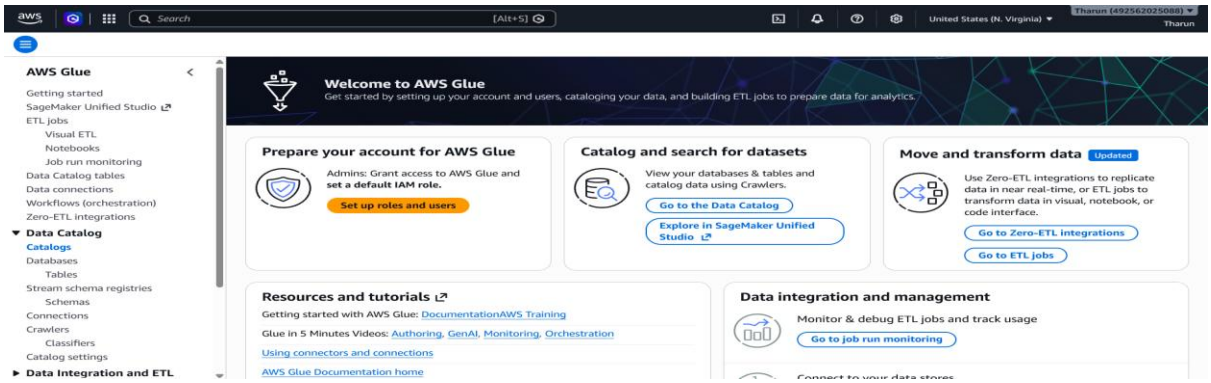
1. IAM Role used for AWS Glue



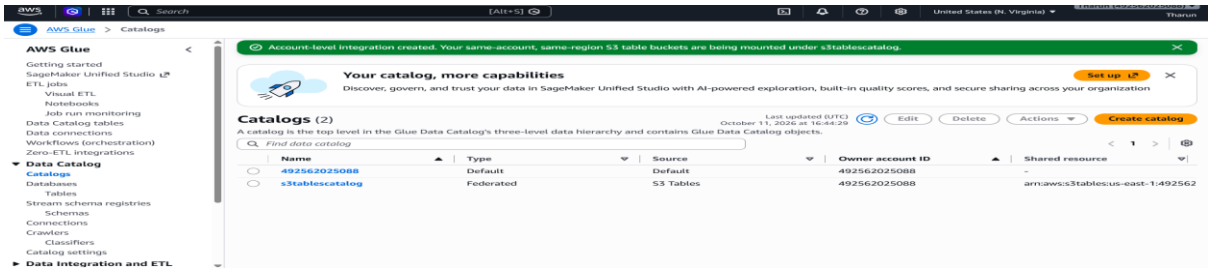


AWS GLUE DATABASE

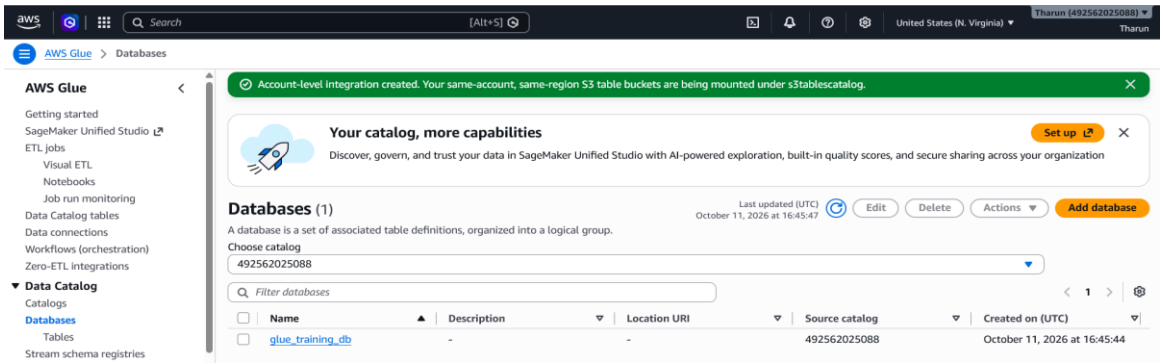
1.AWS Glue Console



2. Data Catalog

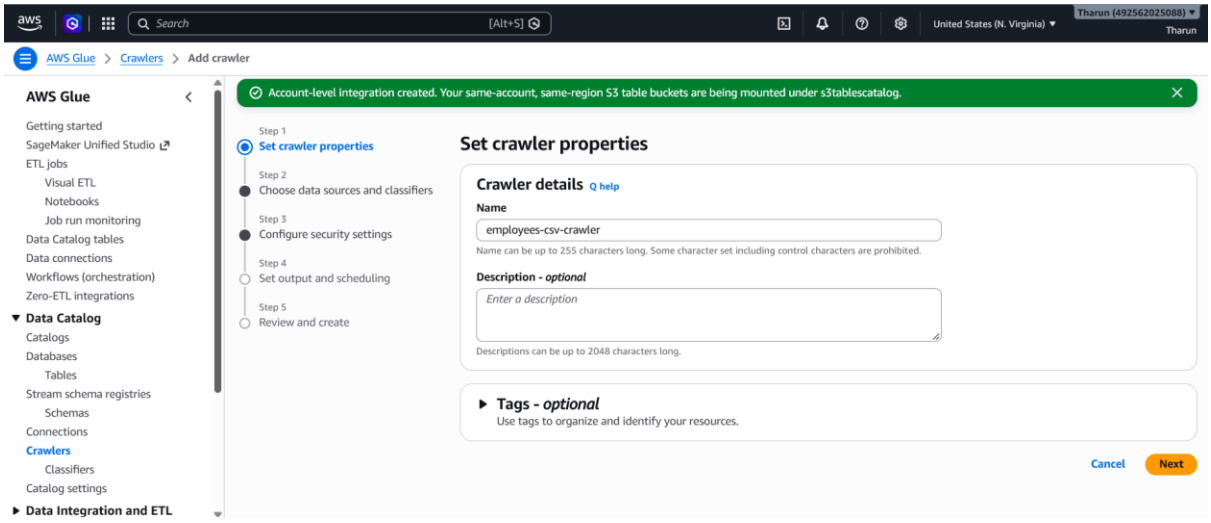


3. Database created

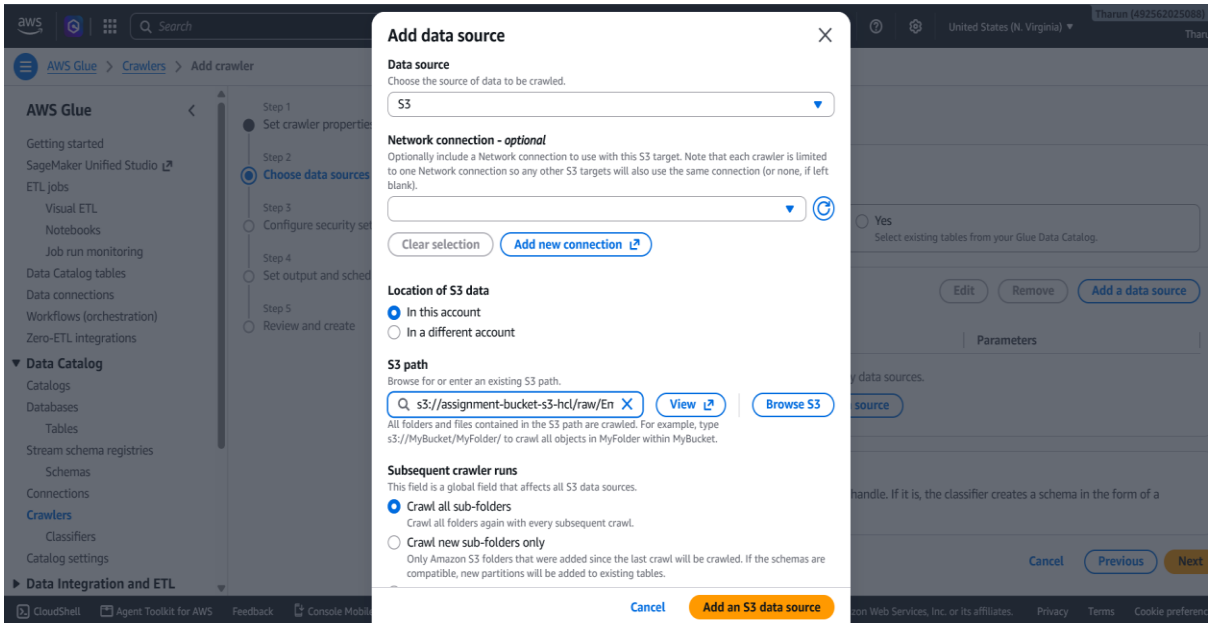


GLUE CRAWLER

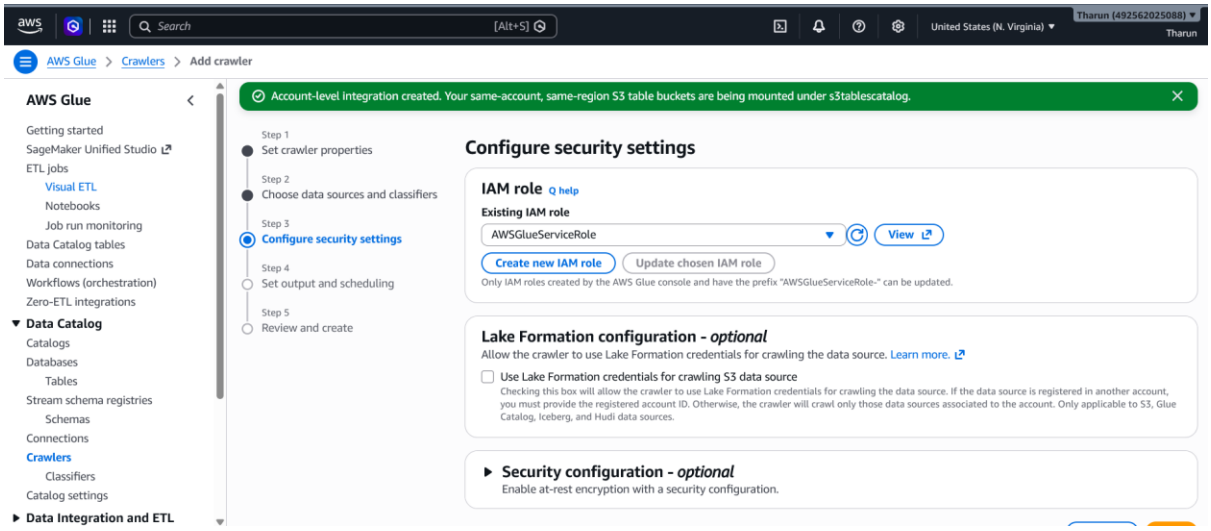
1. Crawler name



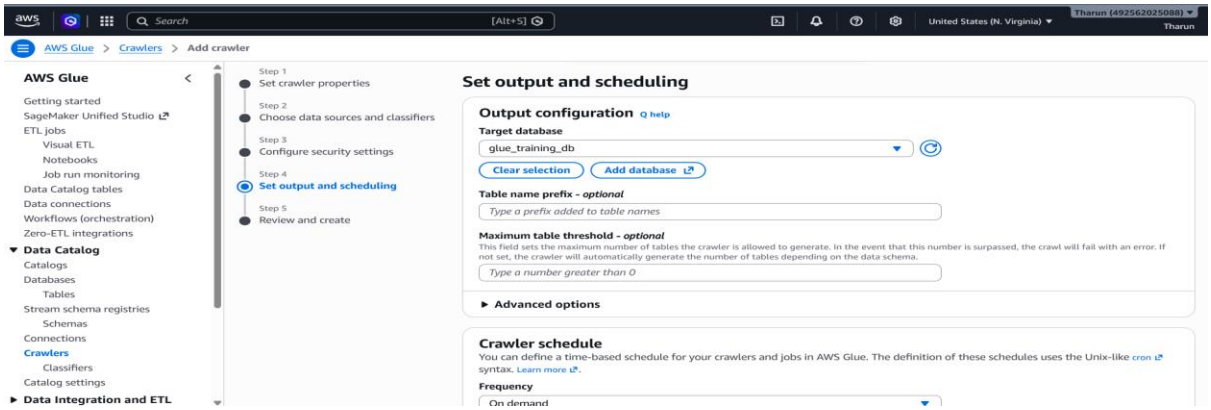
2. S3 source location



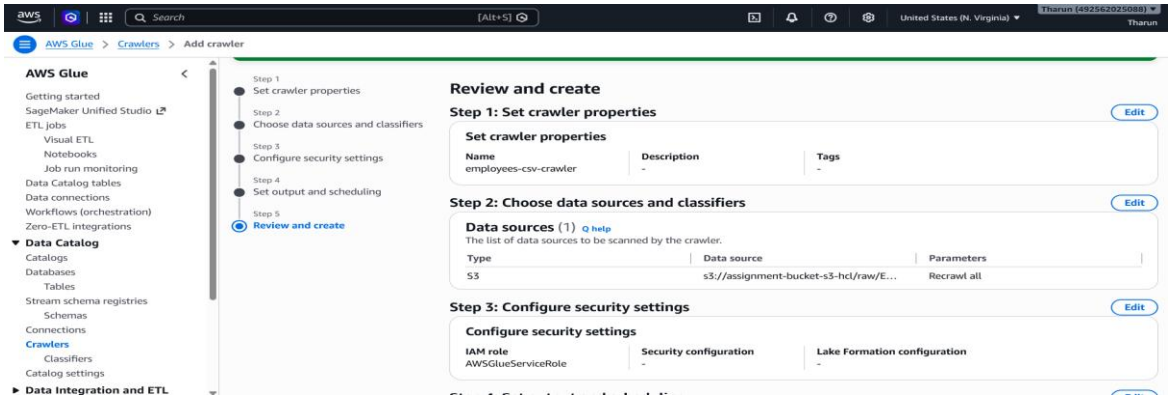
3. IAM role



4. Target database

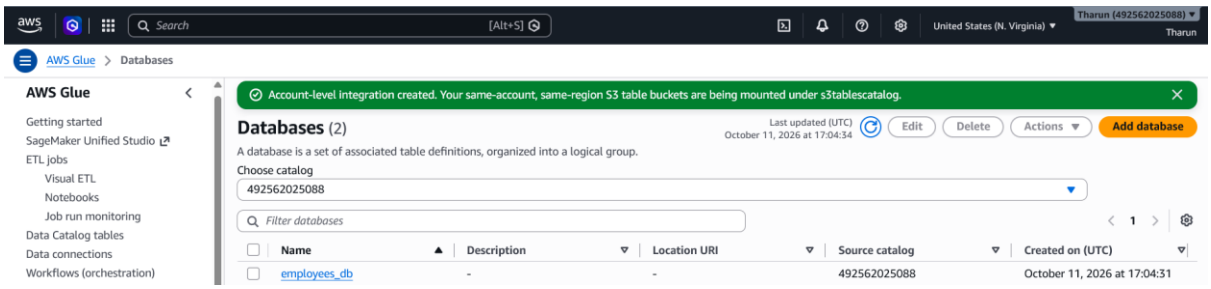


5. Crawler configuration

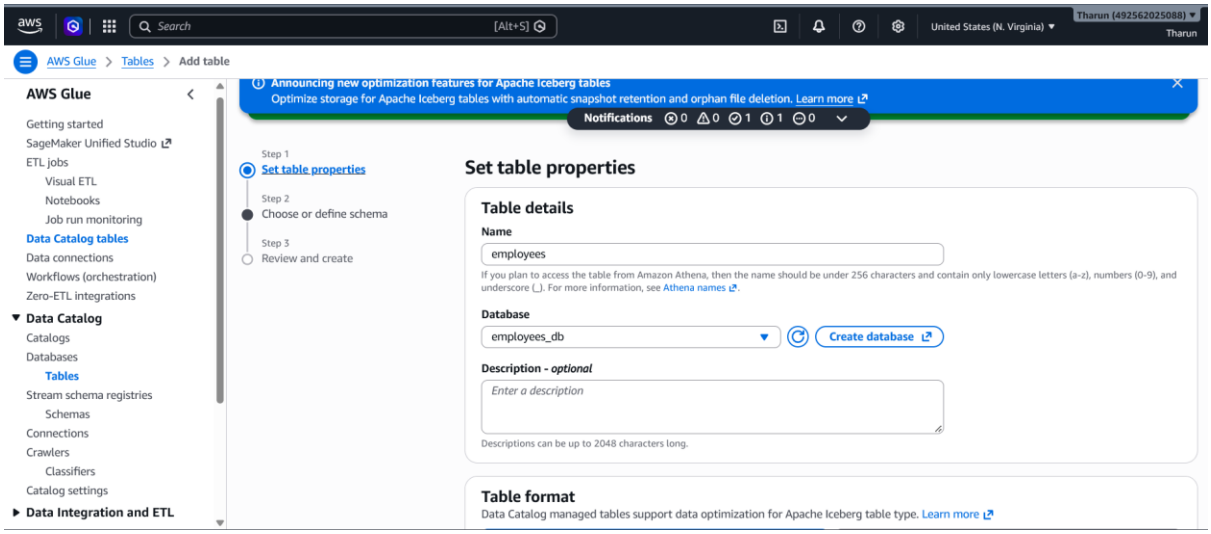


GLUE DATA CATALOG

1. Database



2. Employees table



3. Table successfully created

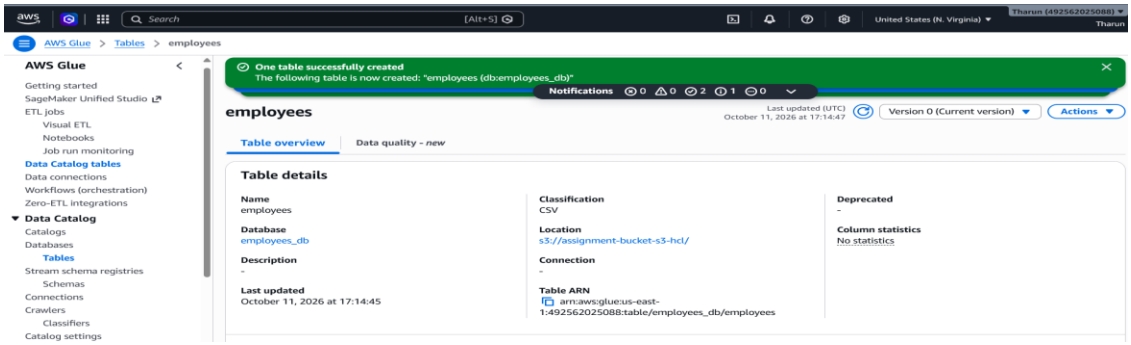
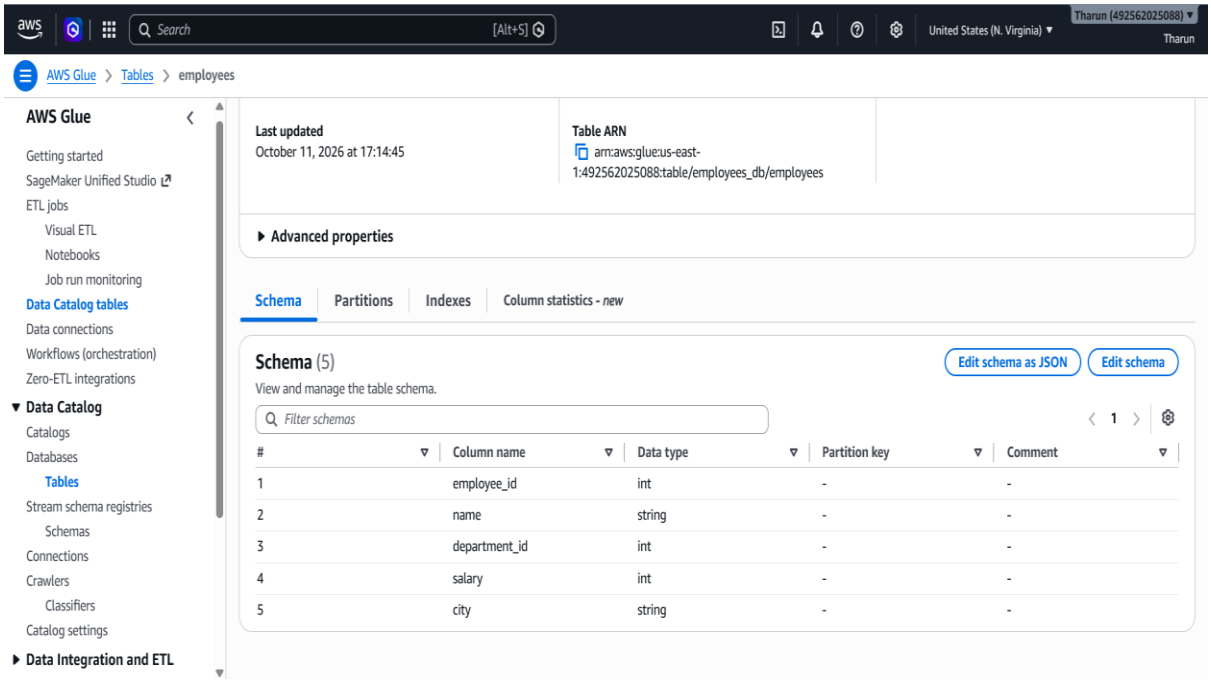
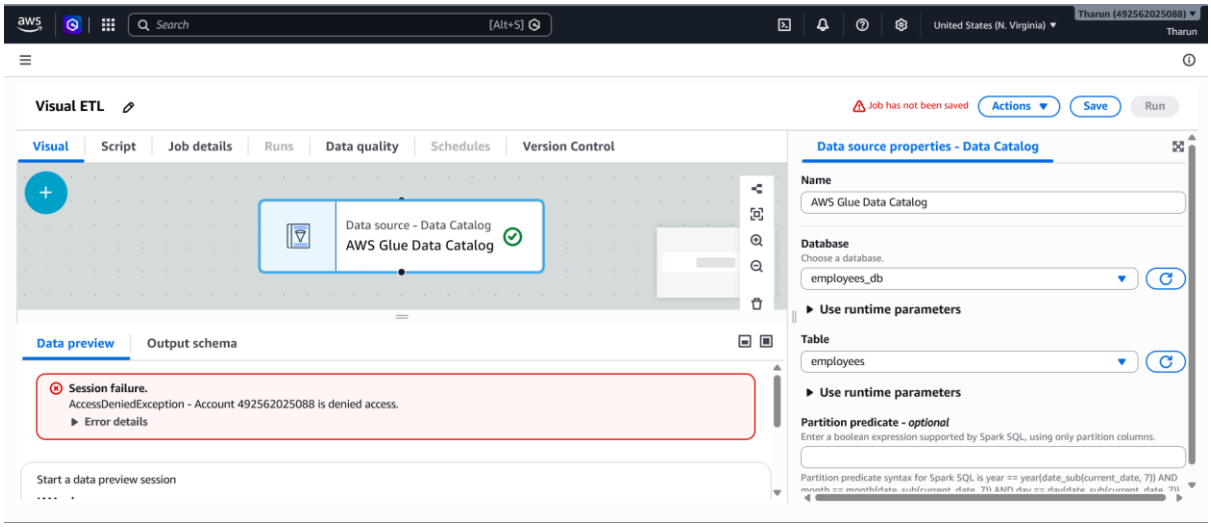


TABLE SCHEMA

- **employee_id**
- **name**
- **department**
- **salary**
- **city**



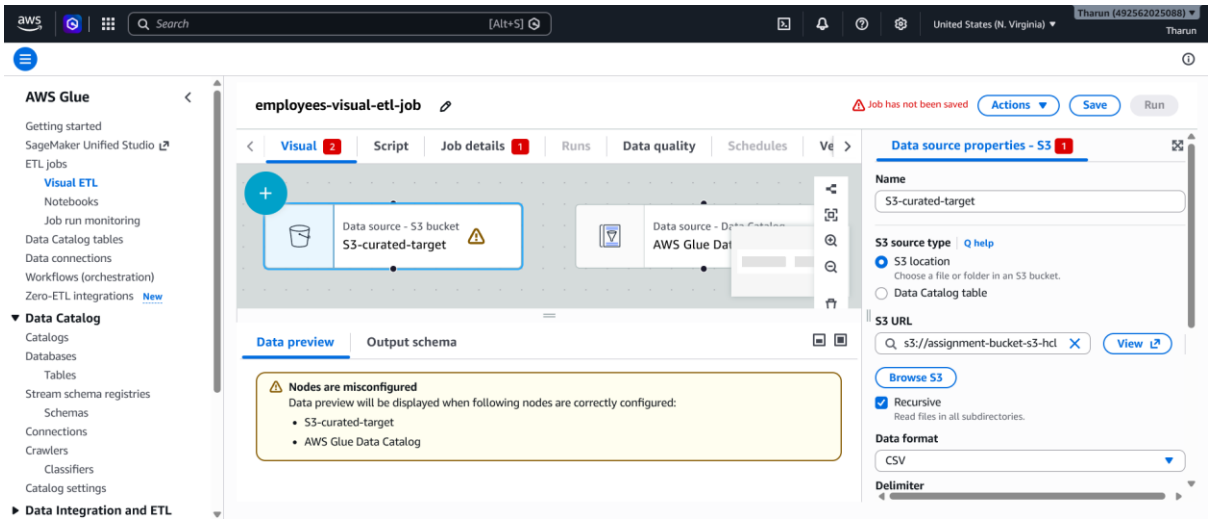
GLUE VISUAL ETL



AWS Glue Visual ETL – Implementation Limitation

The AWS Glue Visual ETL workflow could not be fully validated due to an AccessDeniedException encountered while initiating the Data Preview session. This appears to be an AWS account-level access restriction. As a result, the complete workflow could not be executed and verified in the current environment. The limitation has been documented, and the remaining configuration was attempted to the extent permitted by the available access.

S3 Target Configuration



S3 Target Configuration and Glue ETL Job Execution – Limitation

I was unable to create and execute the AWS Glue ETL job and configure the S3 target due to limitations and resource access restrictions in my current AWS account.

As a result, I could not configure the curated/ output location, select Parquet as the output format, or verify that the job status was SUCCEEDED.

I would require the necessary AWS permissions and resource access to complete these steps successfully.

Final S3 Output – Limitation

I was unable to generate the final S3 output because I could not create and execute the AWS Glue ETL job due to resource access and permission restrictions in my current AWS account.

As a result, I could not verify the creation of the curated/ folder, the generated output files, or the Parquet output format.

The necessary AWS Glue and Amazon S3 permissions and resource access are required to complete and verify this step successfully.

Final Explanation

1. First, I uploaded the employees.csv file to an Amazon S3 bucket, which acts as the storage location for my raw employee data.
2. Next, an AWS Glue Crawler is used to scan the CSV file, identify its structure, and detect the column names and data types.
3. The detected information is stored as metadata in the AWS Glue Data Catalog, where the database and employees table can be managed.
4. AWS Glue Visual ETL is used to design the data processing workflow by connecting the Data Catalog source, transformation step, and S3 target.
5. During transformation, the employee data can be cleaned or modified as required before being stored in the output location.
6. The processed data is intended to be saved in the curated/ folder of Amazon S3 in Parquet format, which is suitable for analytical workloads.
7. Due to resource access and permission restrictions in my AWS account, I could not complete the crawler and ETL job execution, so the final Parquet output could not be verified.